INTEGRATIVE PROGRAMMING AND TECHNOLOGIES

React + Laravel Full-Stack Project Proposal

Grouping

- **5 members per group only**
- All members must participate in frontend, backend, database, testing, deployment, and presentation.

Project Requirement

Each group must develop a **full-stack web application** using:

- **ReactJS** – Frontend
- **Laravel** – Backend/API
- **Database** – MySQL or approved database
- **Cloud Deployment** – Final application must be deployed online

The project must demonstrate proper integration between the React frontend and Laravel backend.

Required Technologies

Frontend

The React application must demonstrate:

- Components
- JSX
- Props
- State
- Event Handling
- React Hooks
- API communication

Backend

Laravel must be responsible for:

- API routes
- Controllers
- Models
- Validation
- Database operations

- Business logic

CRUD Requirement

The system must include complete:

- Create
- Read
- Update
- Delete

CRUD operations must work from the React interface through the Laravel backend.

Additional Features

The system must include at least **two additional functions**, such as:

- Search
- Filter
- Sorting
- Authentication
- Status management
- Categories
- Dashboard
- Reports
- Pagination

Cloud Deployment

The completed project must be deployed online.

Both the frontend and backend must be accessible through the internet.

The group must submit the deployment links during final presentation.

Proposal Content

I. Project Title

Provide a specific title for the web application.

II. Introduction

Write **2–4 paragraphs** explaining:

- the existing problem;
- the intended users;
- why the system is needed; and
- how a web-based solution can help.

III. Statement of the Problem

Provide the main problem and at least **three specific problem questions**.

IV. Objectives

Provide:

- One General Objective
- At least three Specific Objectives

V. Target Users

Identify the users of the proposed system.

VI. Proposed Features

List the main functions of the system.

VII. ReactJS Implementation

Explain how React will be used for:

- UI components
- forms
- state management
- events
- API requests

VIII. Laravel Implementation

Explain how Laravel will be used for:

- backend logic
- API endpoints

- validation
- database operations

IX. Database

Identify the main data that will be stored.

Include the proposed major tables or entities.

X. Cloud Deployment Plan

Identify where the application may be deployed.

The proposal should explain how the group plans to make the project available online.

XI. Expected Output

Describe the final full-stack web application.

Minimum Requirements

The final project must:

1. Use ReactJS.
2. Use Laravel.
3. Use a database.
4. Implement complete CRUD.
5. Include at least two additional functions.
6. Connect React to Laravel using an API.
7. Include input validation.
8. Have a usable interface.
9. Be deployed to the cloud.
10. Be demonstrated live during presentation.